

Spatial point pattern analysis of environmental effects on valley fever intensity in Phoenix, Arizona

John Ginos^a, Beckett Sterner^{b,*}, Petar Jevtić^a

^a School of Mathematical and Statistical Sciences, Arizona State University, 901 Palm Walk, Tempe, 85281, Arizona, USA

^b School of Life Sciences, Arizona State University, 427 East Tyler Mall, Tempe, 85281, Arizona, USA

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ABSTRACT

Using a novel Arizona Medicaid data set, we model Valley fever (VF) cases in the Phoenix Metropolitan Area as a spatial point process during six month intervals between 2013 and 2023. We estimate the intensity function of VF cases, observed at residential locations of patients, as a function of environmental covariates available at high spatial resolutions: exposure to various land cover development categorizes, exposure to changes in land cover type, Normalized Difference Vegetation Index (NDVI), change in NDVI, and a Habitat Suitability Index (HSI) for the *Coccidioides* fungus. All covariate data were spatially smoothed using a Gaussian kernel with a 1-kilometer bandwidth to capture local variation around residential addresses of patients. Models incorporating environmental covariates perform best in 21 of 22 semesters. We show that HSI measured at the residential (point) level is useful for understanding VF risk. Furthermore, we find that NDVI is often negatively associated with the intensity of VF cases. Despite improved model fit when compared to our simplest model, we show that even our best, most complex model consistently underestimates the intensity in portions of the study window. Our results provide initial evidence that spatially localized environmental information is useful for explaining the intensity of VF cases among Medicaid patients in the study window.

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1. Introduction

Coccidioidomycosis, commonly known as Valley fever (VF), is a disease caused by the fungal pathogens *Coccidioides posadasii* and *C. immitis*, which are endemic to North America. The geographic range of Valley fever and number of human cases are growing, especially in the western United States (Howard et al., 2024; Morgan et al., 2019). Identifying environmental factors that influence the number of VF cases in a specified time period (the intensity of VF cases) is important to public health authorities and communities. Among the environmental factors identified in recent work on the intensity of VF cases and seasonal variability are temperature, precipitation, air quality, and soil properties (see (Kollath et al., 2022), (Dobos et al., 2021), and (Weaver & Kolivras, 2018); see also (Porter et al., 2024) for a succinct summary of variables and methods used in the analysis of VF data). Often, these studies aggregate VF cases to administrative geographical units like

* Corresponding author.

E-mail address: bsterner1@asu.edu (B. Sterner).

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census tracts, counties, or cities (e.g. see (Head et al., 2022), (Kollath et al., 2022), and (Porter et al., 2024), respectively). Aggregation of spatially referenced location data to larger geographical units may present problems for data analysis including the modifiable areal unit problem (Buzzelli, 2019), in which results from spatial data analysis are greatly impacted by the scale of aggregation. A related example of this when analyzing VF data specifically is discussed in (Brown et al., 2017), where the authors report that percentages of land cover type in geographical units showed drastic variation depending on the spatial scale of units used (e.g. census tract, block group, and zip code). We seek to avoid this problem to the greatest extent possible by considering the VF case locations (by residence) at the point level and using higher resolution covariate data. However, since exposure to *Coccidioides* arthroconidia is not limited to one's residential location, we simultaneously explore the utility of spatially local covariate information (i.e. information around one's residence) in understanding VF risk.

To these ends, we employ the first spatial point process model¹ of residential addresses of human VF cases based on dataset of Arizona Medicaid patients.² The dataset includes VF cases of Medicaid patients in and around the Phoenix Metropolitan Area during 2013 to 2023 (hereafter referred to as the study period). In 2023, Maricopa County, Arizona, which includes the greater part of the Phoenix Metropolitan Area accounted for roughly 38% of national VF cases according to provisional data from the Centers for Disease Control and Prevention (CDC) as well as figures reported by Maricopa county; see (Centers for Disease Control and Prevention) and (Maricopa County Department of Public Health, 2025).

While one-dimensional temporal point process models have been employed to understand the intensity of VF cases over time (Wang et al., 2021), we are not aware of any formal treatment of VF cases as a spatial point pattern. The most granular analysis of VF data in the literature is the logistic regression approach in (Brown et al., 2017), where the authors used presence/absence data derived from residential addresses of those with VF and pseudo-absence data randomly generated based on the population of census block groups in Arizona. A comparison of spatial point pattern analytic methods with the logistic regression approach (using pseudo-absences) similar to the one used in (Brown et al., 2017) and the limitations of this latter approach are discussed in (Warton & Shepherd, 2010) in the case of ecological data analysis. We consider environmental factors at all locations where individuals could live in the study window at the finer spatial scale. To accomplish this, we treat the residential addresses of VF cases as a spatial point pattern and employ well-established techniques³ to analyze factors that influence the spatially inhomogeneous intensity of VF cases. This approach also allows us to obtain fitted intensities for the whole study window (and at the locations of VF cases themselves), which can be used to determine where the model fit breaks down.

We find evidence supporting spatially localized effects of environmental covariates, including the degree of local exposure (of a residence) to high-intensity (HI), medium-intensity (MI), low-intensity (LI) and open development land cover categories, vegetation level (NDVI), and soil habitat suitability for *Coccidioides* fungal pathogens in most semesters of the study period. The support we find for biotic spatial variables such as vegetation level and change is consistent with the enzoonotic hypothesis that rodents play an important role in spreading the disease, although we are not able to directly investigate the importance of wind versus animal-borne dissemination mechanisms as discussed in (Porter et al., 2024).

While the Medicaid dataset we analyze has several limitations, our results suggest that VF exposure is associated with environmental covariates at a fine spatial scale and that additional environmental variation and/or nonlinear model structure may explain residual spatial clustering we observe in the data based on our best model.

2. Data and methods

2.1. Valley fever data

Locations of residence for individuals diagnosed, treated, and/or tested for VF were obtained from the Arizona Health Care Containment System (AHCCCS; the Arizona state Medicaid agency) during the study period 2013 through 2023. The Arizona Medicaid (AHCCCS)⁴ data used in this study were provided by the Arizona State University, Center for Health Information Research (CHiR). We received approval from the Arizona State University Internal Review Board (Study 20039) prior to the beginning of research and followed all HIPAA protocols to protect human subjects data. The data were aggregated by semester⁵ and year with the first semester of each year defined as January 1st through June 30th; the remaining days of the year constitute the second semester. All Medicaid enrollees represented in the data set were identified with a member identification code associated with unique patients who were continuously enrolled during a given semester.

Demographic information for each patient including age and sex were included along with whether the individual was tested, treated, and/or diagnosed with VF in a given semester. Only the earliest diagnosis of VF was considered for patients with multiple (non-consecutive) diagnoses to avoid double counting unique diagnoses for individuals who may have

¹ All code, covariate data and simulated VF data are available at <https://github.com/jginos/VF-SPP-Analysis>.

² We gratefully acknowledge Arizona State University's Center for Health Information Research (CHiR) and the Arizona Healthcare Cost Containment System (AHCCCS) for access to this data.

³ See chapter 9 of (Baddeley et al., 2015) and section 2.3.

⁴ Hereafter, we refer to this data as Arizona Medicaid.

⁵ The semesterly time series of VF cases for the Medicaid population in Maricopa County had a 79% correlation with the series of all VF cases for Maricopa County as reported by the Arizona Department of Health Services (with 2023 data being considered provisional at the time of the analysis; see (Arizona Department of Health Services, 2024) for annual reports on Valley fever Cases in Arizona Counties).

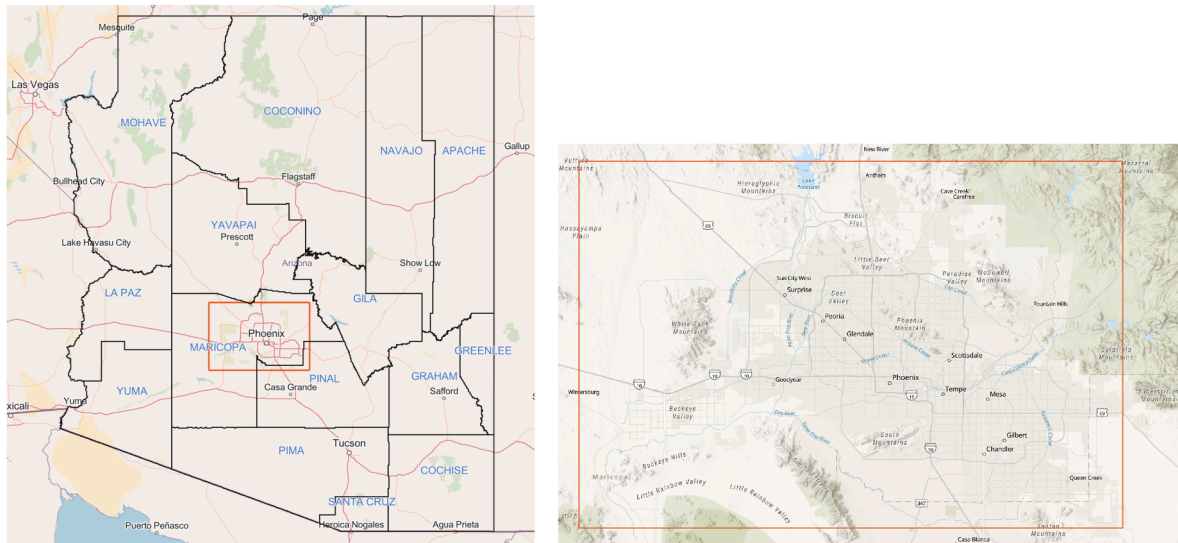


Fig. 1. Left: Bounding box of the CAP LTER (red box; this is roughly the area of the study window) as situated within Arizona. Arizona county borders (black) are shown along with county names in blue. This image was taken as a screenshot of the OpenStreetMap basemap available in ArcGIS Online ([OpenStreetMap contributors, 2025](#)). County boundaries obtained from ([Arizona Geographic Information Council, 2025](#)) and ([ArcGIS Online Contributor, 2025](#)). Right: Bounding box of the CAP LTER study window (taken as a screenshot of the topographic basemap available in ArcGIS Online ([Esri](#))).

unenrolled and re-enrolled in Medicaid during the study period. The demographic information was used to categorize the VF cases into four groups based on age and sex (see 2.3 for more details). Splitting the locations of VF cases into sub-patterns of spatial points allows us to consider how each pattern's spatial distribution is influenced by the same set of covariates and account for known heterogeneity in VF prevalence among such groups.

Due to the extreme contrast in the spatial density of VF cases between major Arizona metropolitan areas (like Phoenix and Tucson) and rural regions of the state and due to covariate data availability, we decided to limit our analysis to an area roughly equivalent to the Central Arizona-Phoenix Long-Term Ecological Research study region within metropolitan Phoenix⁶ (hereafter study window; see [Fig. 1](#)).

2.2. Spatial covariate data

As mentioned above, we are interested in how the intensity of VF cases is influenced by environmental covariates including the soil Habitat Suitability Index (HSI) of the *Coccidioides* fungus (see ([Dobos et al., 2021](#))), NDVI, land cover category, and changes in land cover category that occur from one year to the next (we call this land cover change). Certain land cover changes are considered relevant for our purposes (see 2.2.5). Aside from environmental covariates, we also use the estimated Medicaid population density to account for the spatially varying population at risk within the study window.

All covariate data were rasterized⁷ (if not already in raster format) and converted to a pixel image object for compatibility with the R package *spatstat* ([Baddeley & Turner, 2005](#)).

One of our primary objectives is to understand how environmental data recorded at the smallest available spatial scales informs the spatially varying (inhomogeneous) intensity function of the underlying process of VF cases. Specifically, we want to consider both how environmental data recorded at and around the residence of the patients effect the intensity of VF cases. Moreover, we assume that neighboring environmental effects decay rapidly in all directions with distance. Thus, after appropriate processing, all covariate data were smoothed using a Gaussian kernel with a 1-kilometer bandwidth. This includes a smoothed, real-valued version of the land cover and land cover change variables that would otherwise be categorical-type variables (see 2.2.4 and 2.2.5).

⁶ While the CAP LTER is technically not a perfectly rectangular region, We have chosen to use the area limited by the NDVI rasters obtained for this analysis and which the authors who made the data available at ([Stuhlmacher et al., 2013](#)) have designated as the CAP LTER study region. The study window is roughly the bounding box of the CAP LTER study region. After projecting then NDVI rasters, (using the WGS 84/Pseudo-Mercator projection) the resulting area deviated from the original rectangular shape.

⁷ All rasters, including those for land cover, NDVI, land cover change, and HSI are projected using the WGS84/Pseudo Mercator projection and rescaled to reflect measurements in kilometers. NDVI was projected using the cubic spline method available as an option when using the R package *Terra*. Land cover data are projected using the modal pixel resampling scheme while land cover change was projected using the nearest neighbor scheme.

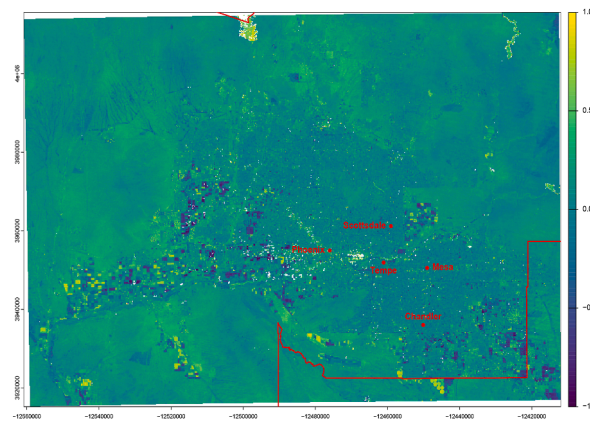


Fig. 2. Normalized difference (by pixel) between fall NDVI of 2023 and the fall NDVI of 2012 (the year prior to the start of the study period; 2012 NDVI data were used for creating a differenced NDVI covariate). Bright yellow areas denote an increase in NDVI over this period while dark purple areas represent a decrease in NDVI. Blank spaces are NA. Phoenix and a few nearby city centroids labeled in red along with county borders. **Source:** The NDVI data used to create this figure was obtained from (Stuhlmacher et al., 2013).

2.2.1. Medicaid population density (MPD) estimates

We expect that the spatial distribution of VF cases varies with the Medicaid enrollment population density (MPD), which varies differently than the overall population density due, in part, to socioeconomic factors. Since we did not have access to the true MPD, we used the [U.S. Census Bureau's](#) American Community Survey (ACS) 5-year estimates for Medicaid enrollment for the study period⁸. Estimates are available at the census block group level; however, when aggregated to the county level, they underestimate the true Medicaid enrollment figures available from AHCCCS during the study period. To address this, we normalized all block group estimates (for all of Arizona) to sum to their respective county-level estimates⁹ with data obtained for AHCCCS (for monthly AHCCCS enrollment data see ([Arizona Health Care Cost Containment System AHCCCS, 2024](#))).

Areas where the Medicaid enrollment population is estimated to be zero present a numerical problem when modeling spatial point data using an offset to account for spatially varying exposure. To avoid zero-MPD areas, we set these block group estimates to one-fourth of the margin of error pertaining to the estimate¹⁰. The resulting Medicaid enrollment block group estimates are divided by block group area (in square miles) to obtain MPD.

To further reduce potential numerical issues related to outliers and to deal with the uncertainty in the estimates themselves, we rasterize (to 1-kilometer resolution) and then smooth the MPD block group estimates using a Gaussian kernel. The bandwidth of the Gaussian kernel is chosen to be 2 kilometers to preserve local features in the MPD estimates but avoid large outliers in the population estimates. Finally, we normalize the smoothed MPD estimates (SMPDE) for use as an offset term in modeling VF cases. This is done by dividing all resulting pixel values by the sum of all pixel values in the study window. Such normalization reduces the remaining large-scale spatial variation in the offset term to avoid numerical problems when model fitting while maintaining positive values at each pixel location.

2.2.2. Normalized Difference Vegetation Index (NDVI)

Raster data for the study window were obtained at the 30-meter resolution from the Environmental Data Initiative (Stuhlmacher et al., 2013). Seasonal NDVI rasters for the study period and the year 2012 were used for the analysis. The average NDVI (by raster cell/pixel value) for winter and spring¹¹ of a given year was used for semester 1 covariates while the average NDVI for the summer and fall were used as semester 2 covariates during the study period.

NDVI is known have a lagged positive effect on VF cases for up to and slightly beyond a year (see (Stacy et al., 2012)), though this lagged relationship has not been explored at the finer spatial scale of interest here. Unfortunately, the VF case data does not align with the seasonal NDVI data; it is hard to link our case data with seasonal variations in NDVI since individuals in the data set could have been diagnosed or treated at any time in a six month period. For simplicity, we explore

⁸ Estimates regarding healthcare coverage by demographic group and at the block group level can be found ACS tables with ID B27010. This was used in conjunction with 2019 and 2022 TIGER/Line block group shapefiles provided by the U.S. Census Bureau to represent the 2010 and 2020 block group geometries (U.S. Census Bureau, 2019), (U.S. Census Bureau, 2022), (U.S. Census Bureau).

⁹ This was done for all census block groups in Arizona but only the subset of block groups contained in the study window were included in the analysis (this includes fractional portions of block groups near the boundaries of the study window).

¹⁰ This fraction does not significantly impact the analysis as block groups with an estimated Medicaid enrollment population of zero not likely locations for VF cases and did not have large margins of error.

¹¹ Winter rasters include landsat imagery from December 21st the prior year through March 19th of the current year. Spring rasters include imagery from March 20th through June 20th and Fall NDVI rasters include imagery starting on June 21st.

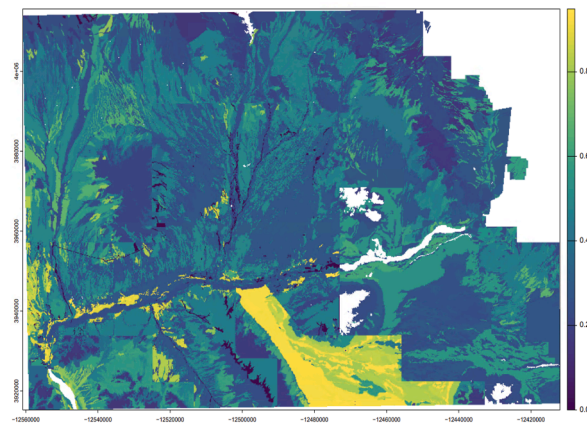


Fig. 3. *Coccidioides* habitat suitability (HSI) index shown at the 10-meter resolution within the study window. Blank spaces are NA. **Source:** Obtained with the help of Robert R. Dobos and links to the original data sources provided in (Dobos et al., 2021).

the NDVI and lagged NDVI effects up to one semester prior. Once the NDVI averages were obtained for each semester, a *differenced* NDVI covariate was obtained by differencing the NDVI for each semester with the NDVI from the prior semester. For example, for 2013 semester 1, the differenced NDVI covariate added into the model by computing the pixel-wise difference between the winter-spring average NDVI for 2013 and the fall-summer average NDVI for 2012.¹² Fig. 2 shows the normalized difference of fall NDVI rasters from 2012 to 2023. The normalized difference, given by $\frac{NDVI_{2023} - NDVI_{2012}}{NDVI_{2023} + NDVI_{2012}}$, provides a useful pixel-wise calculation for inspecting changes from the semester just prior to and near the end of the study period. Note the significant areas of change in the west and southeastern portions of the study window. The increase in population and urban development in these areas may explain the apparent decreases in NDVI.

2.2.3. *Coccidioides* habitat suitability index (HSI)

Recent work in (Dobos et al., 2021) showed that certain soil characteristics provide suitable conditions for the *Coccidioides* fungus. The authors¹³ in (Dobos et al., 2021) created the Habitat Suitability Index to provide a spatially granular method of measuring where exposure to arthroconidia is most likely. The index takes values between 0 and 1 with values closer to 1 indicating increased suitability. The index is, in part, derived from available soil survey and climatic data. A resulting 10-meter resolution raster is obtained from the authors' predicted habitat suitability index within the study window.¹⁴ Fig. 3 shows the HSI raster for the study window; the large block of NA pixels¹⁵ in the northeast corner is due to unavailable soil survey data for the Tonto National Forest. Note that we use a single HSI covariate raster for the entire study period (since the HSI data does not vary significantly by year and semester).

2.2.4. Land cover category

The Multi-Resolution Land Characteristics (MRLC) Consortium provides data from the National Land Cover Database (NLCD) for use by the public; see (Geological Survey). From the available data, we have obtained annual land cover characteristics at the 30-meter resolution for the study window for 2013 through 2023.

The land cover types we include in our analysis indicate the degree of development (constructed materials on top of natural soil (Multi-Resolution Land Characteristics (MRLC) Consortium, n.d.)), which is directly related to the amount of potentially uncovered soil that may contain the *Coccidioides* fungus. Uncovered soil is more likely to be disturbed in a way that could lead to the ambient dispersal of arthroconidia. We seek to understand how the degree of exposure to uncovered soil affects the intensity of VF cases. We note, however, that land cover type is also likely associated with different types of human activities as well, and so cannot be simply interpreted as an ecological factor only. To construct covariates that reflect the degree of exposure to uncovered soil we transform the raster of land cover categories into multiple real-valued rasters that give a measure of the proximity and area classified according to various land cover development categories. This is done by first creating binary rasters for each of four development categories (classifying 30-meter pixels based on land cover development category) followed by a smoothing of the result using a Gaussian kernel with a 1-kilometer bandwidth. Thus,

¹² Differenced NDVI covariates were obtained after conversion from a raster to a pixel image for compatibility with the R package *spatstat*.

¹³ We would like to thank Robert R. Dobos for help with obtaining the HSI data for this study.

¹⁴ Using gridded Soil Survey Geographic Database (SSURGO) data and a table with HSI values containing a map unit key, the HSI raster layer was created using ArcGIS Pro 3.3.2 (Esri, 2024).

¹⁵ However, it must be noted that for this study, some areas (that are relatively small in size) that appear marked as 0 (dark purple) should in fact be NA but were labeled incorrectly. We have adjusted many of these pixels manually. Moreover, we are confident that areas marked as 0 that should be marked NA do not impact the analysis significantly.

pixels in the study window are assigned a weighted average based on their proximity to pixels where the land cover category was observed.

We are interested in understanding how the degree of exposure of a particular location to any of four land cover development categories is associated with the intensity of VF cases at that location. The categories we consider are open-area development,¹⁶ low-intensity development, medium-intensity development, and high-intensity development.

As previously mentioned, the Medicaid population density data contains inherent uncertainty. We expect that the degree of exposure to high and medium-development land cover will, to some degree, reflect the spatially varying population density within the study window. All else being equal, we expect that VF cases are more likely to occur near densely populated areas which are generally associated with greater urban development (i.e. greater exposure to high and medium-development land cover). We also expect that areas with less development (especially on soils with a high HSI) are more suitable habitats for animal hosts of *Coccidioides*, so that individuals with greater exposure to low-intensity and open development may be at greater risk of VF exposure. Thus, we hypothesize a positive association between the intensity of VF cases and greater degrees of exposure to each of the four land cover development categories under consideration.

2.2.5. Land cover change

Construction, land development and renovation of existing structures often involve soil-disturbing activities like excavation. Displacing soil may cause increased risk for VF among both construction workers and those who live near the development site (Laws et al., 2018; Wilken et al., 2015). To consider the effect that such development sites may have on the intensity of VF cases within the target population, we use land cover change data from the NLCD made available by the MRLC. The raster data classify 30-meter pixels in the study window based on whether a change from one land cover category to another occurred from one product year to another (product years are when imagery is produced for public use).

Recent work in (Porter et al., 2024) investigated the effects of land cover change (mostly increased urbanization), among other factors, on the presence of *Coccidioides* arthroconidia in air filters placed throughout the Phoenix Metro area. Their approach was to consider the area within a 2.4-kilometer radius of air filter sites and calculate the proportion of area that underwent any land cover change. Note that the authors included all possible changes of land cover types in the NLCD raster data. The authors found no major relationship between air prevalence of *Coccidioides* arthroconidia and land cover change. Similarly, the relationship between human clinical cases and air prevalence of *Coccidioides* arthroconidia was also insignificant in that study.

We study the relationship between our subset of disaggregated case data and certain classes of land cover change. To properly capture land cover change, we create a 0 – 1 binary raster¹⁷ indicating pixels with a 1 if, from the previous year to the particular year under consideration, a site experienced a change from a non-development category to a development category (i.e. the land cover development categories described in 2.2.4) and a 0 otherwise. We then smooth the result as done with the other covariates; smoothing is particularly useful here as many individuals can live near multiple sources of land use change and the resulting smoothed covariate surface acts as a measure of how exposed particular locations are to soil disturbing activities associated with new development. We hypothesize that land cover change to more developed categories would be positively associated with intensity if it was associated with disturbance of arthroconidia from the soil to the air where human exposure can occur.

2.3. Methods: spatial point pattern analysis

We seek to estimate the intensity function of the VF cases over the study window as a function of high-resolution environmental covariate data. Loosely speaking, the intensity function of a spatial point process can be used as a measure of the expected number of cases in any bounded subregion of the study window W (including the entirety of W itself; see (Baddeley et al., 2015), chapter 6, section 6.3 for specifics on the intensity function of spatial point processes). For individuals infected with VF, it may be difficult or impossible to identify the exact spatial location of the exposure that caused the infection. However, the locations of residence of patients infected with VF may offer valuable insight into the spatio-temporal dynamics of the disease. For example, in locations where soil and other environmental conditions are right, nearby soil disturbance due to new construction, dust storms, or other activity may infect multiple individuals in close proximity. Analyzing residential locations as a spatial point pattern allows us to understand the local conditions that increase the likelihood of VF exposure. Among the benefits of this approach is being able to extract covariate effects on disease intensity at a much smaller scale than if data were aggregated, while avoiding the so-called modifiable areal unit problem inherent in spatial data analysis. Treating VF cases as a spatial point pattern allows us to leverage environmental data collected at much finer spatial resolutions, which is especially useful when the spatial distribution of VF cases exhibits high levels of inhomogeneity over relatively small spatial scales (e.g. kilometers).

¹⁶ According to the NLCD class legend and description " ... These areas most commonly include large-lot single-family housing units, parks, golf courses, and vegetation planted in developed settings for recreation, erosion control, or aesthetic purposes." (Multi-Resolution Land Characteristics (MRLC) and Consortium).

¹⁷ When projecting land cover change rasters to the WGS 84 coordinate system, a nearest neighbor scheme was used for sampling pixel values.

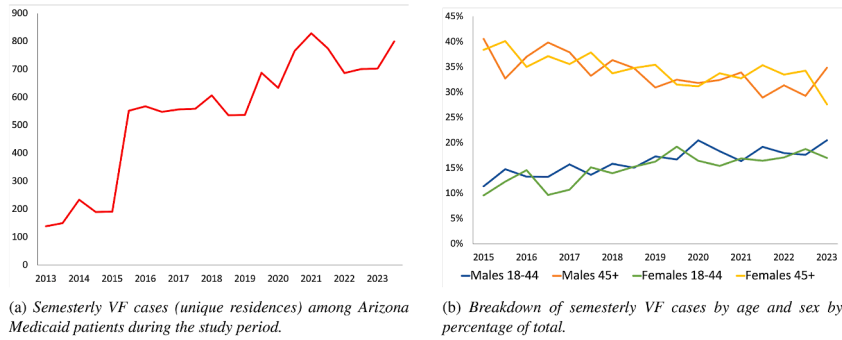


Fig. 4. Source: AHCCCS 2023. CHIR created patient record files from Arizona Medicaid (AHCCCS) data that were used for this analysis.

Our approach involves a 250×250 grid of quadrature points over the entire study window, which corresponds to extracting covariate and VF case information in a grid with horizontal and vertical spacing of approximately 0.6×0.4 km (i.e. this is the spacing of the quadrature points in the horizontal and vertical directions, respectively). This differs from the approach in (Brown et al., 2017), where the authors construct the number and location of pseudo-absences based on census block group information in their study window (the entire state of Arizona).¹⁸

By analyzing the VF cases at the point level on the scale of kilometers, we may construct a smoothly varying surface of the intensity of VF cases. Specifically, we treat the VF cases as a spatially inhomogeneous, ‘multitype’ Poisson process. A ‘multitype’ process in our case refers to the categorization of VF cases based on qualitative attributes (or types) of the points called marks; we analyze the pattern as a multivariate spatial point process. In our case, each VF case may have one of four mark categories based on the age and sex of the person infected. The number of Arizona Medicaid patients during each semester of the study period are shown in Fig. 4 and Table 1.

The age categories were determined by considering the distribution of VF cases by age as given in the Arizona Department of Health Services ‘Valley Fever Annual Report’ (Arizona Department of Health Services, 2023) over multiple years. It should also be noted that Medicare eligibility, which begins at age 65, likely affects the Medicaid enrollment populations among older individuals.

Duplicate points, where both the mark category and location coincide (e.g. in the case of multiple people infected with VF in the same home), cause a technical violation of the assumption of a spatial Poisson process. Thus, to deal duplicated points, we randomly shuffle the indices of the entire data set and take only the first mark category appearing in the shuffled data set at each location. The result is the number of unique addresses where VF cases were observed in each semester/year.

The MPD estimates vary dramatically within the study window, and it is well known that environmental factors influence the abundance of VF cases in Maricopa County (Kollath et al., 2022). Thus, we model the spatially inhomogeneous intensity of VF cases as a function of environmental covariates. Furthermore, since VF is not communicable among humans, the independence assumption that follows from modeling the VF case as a spatial Poisson process (accounting for the spatially inhomogeneous intensity) is a reasonable one.

For ease of comparison among fitted model coefficients, we standardize the HSI and NDVI covariates by subtracting the mean pixel value and dividing by the standard deviation of pixel values within the study window. Additionally, to account for uncertainty in the underlying MPD estimates and as a method of facilitating model fitting, we smooth the MPD estimates (SMPD) over the study window using a Gaussian kernel with standard deviation equal to 2 km. We then normalize the SMPD by dividing each pixel by the sum over all pixels in the study window for numerical stability when fitting the models.

For year $i = 2013, \dots, 2023$ and semester $j = 1, 2$, we specify log-intensity of VF cases for age-sex category $k = 1, \dots, 4$, denoted by $\log \lambda_{ij}(u, \theta, k)$, at spatial location $u \in W$ as a linear function of environmental covariates. Here we let $\theta := (\gamma_{ij}, \alpha_{ij}(k), \beta_{ij})$ where β_{ij} is a vector of model coefficients with length depending on the number of covariates included in the model and $\alpha_{ij}(k)$ is the age-sex specific intercept term that depends on k . An age-sex specific intercept term embeds the assumption that baseline intensities of VF cases for the various age-sex categories differ from one another. This is in contrast to the assumption that the age and sex related effects are also covariate specific, an assumption we do not adopt for sake of parsimony.

We first consider a log-linear base model with only an offset term and age-sex-specific intercept terms based on the marks of the VF cases. This model takes the form

$$\log \lambda_{ij}(u, \theta, k) = \gamma_{ij} + \alpha_{ij}(k) + \log \text{SMPD}_i(u), \quad (\text{M1})$$

¹⁸ However, it should be noted that the logistic regression approach used in (Brown et al., 2017) would yield similar results to the spatial point process approach as the number of pseudo-absences, either placed on a regular grid or in a uniformly random way over the study window, goes to infinity (see (Warton & Shepherd, 2010) for details on this).

Table 1

Number of unique residential addresses of Arizona Medicaid enrollees diagnosed with and/or treated for VF (VF cases) by age and sex. In parenthesis: The same case counts after removal of points where any of the covariate data were NA (i.e. used for model fitting). Categorical breakdowns in early semesters are omitted to protect privacy. **Source:** Arizona Medicaid (AHCCCS) 2023. CHiR created patient record files from Arizona Medicaid (AHCCCS) data that were used for this analysis.

Year & Sem.	Males 18–44	Males 45+	Females 18–44	Females 45+	Total
2013 1	-	-	-	-	139 (139)
2013 2	-	-	-	-	151 (150)
2014 1	-	-	-	-	240 (234)
2014 2	-	-	-	-	193 (190)
2015 1	-	-	-	-	193 (191)
2015 2	64 (63)	228 (224)	55 (53)	213 (212)	560 (552)
2016 1	86 (84)	190 (186)	74 (70)	231 (228)	581 (568)
2016 2	73 (73)	203 (203)	80 (80)	194 (192)	550 (548)
2017 1	76 (74)	226 (222)	54 (54)	209 (207)	565 (557)
2017 2	88 (88)	212 (212)	62 (60)	201 (199)	563 (559)
2018 1	85 (83)	203 (202)	92 (92)	235 (230)	615 (607)
2018 2	85 (85)	199 (195)	75 (75)	183 (181)	542 (536)
2019 1	81 (81)	187 (187)	83 (82)	191 (187)	542 (537)
2019 2	121 (119)	214 (213)	113 (112)	245 (244)	693 (688)
2020 1	108 (106)	208 (206)	124 (122)	205 (200)	645 (634)
2020 2	159 (157)	249 (244)	126 (126)	240 (239)	774 (766)
2021 1	154 (152)	271 (269)	129 (128)	283 (280)	837 (829)
2021 2	128 (127)	264 (263)	134 (131)	258 (254)	784 (775)
2022 1	132 (132)	203 (199)	114 (113)	248 (243)	697 (687)
2022 2	127 (126)	222 (220)	124 (120)	238 (235)	711 (701)
2023 1	125 (124)	207 (206)	133 (132)	244 (241)	709 (703)
2023 2	166 (164)	280 (279)	139 (136)	226 (221)	811 (800)

where we once again emphasize that u represents a spatial location in the study window W .

To understand the effect of covariates that we have established a priori may have a relationship with VF, we add linear terms for these covariates to the structure of model M1 to get

$$\begin{aligned}
 \log \lambda_{i,j}(u, \theta, k) &= \gamma_{i,j} + \alpha_{i,j}(k) + \log \text{SMPD}_i(u) \\
 &+ \beta_{i,j,1} \text{HighDev}_i(u) + \beta_{i,j,2} \text{Med.Dev}_i(u) \\
 &+ \beta_{i,j,3} \text{LowDev}_i(u) + \beta_{i,j,4} \text{OpenDev}_i(u) \\
 &+ \beta_{i,j,5} \text{LCH}_i(u) + \beta_{i,j,6} \text{HSI}(u) \\
 &+ \beta_{i,j,7} \left(\text{NDVI}_{i,j}(u) - \text{NDVI}_{\text{prev.sem}}(u) \right) + \beta_{i,j,8} \text{NDVI}_{i,j}(u)
 \end{aligned} \tag{M2}$$

where $\text{NDVI}_{\text{prev.sem}}(u) := \text{NDVI}_{i-1,j+1}(u)$ if $j = 1$ and $\text{NDVI}_{\text{prev.sem}}(u) := \text{NDVI}_{i,j-1}(u)$ if $j = 2$. Note that for the NDVI difference covariate in 2013, semester 1, we also use the summer-fall average NDVI for 2012, one year before the start of the study period.

Preliminary nonparametric marginal analysis of the relationship between the intensity of VF cases and the covariates revealed nonlinear (and often approximately quadratic) relationships between most covariates and the intensity of VF cases.¹⁹ To account for simple nonlinear structure, we consider a log-quadratic model of intensity given by

$$\begin{aligned}
 \log \lambda_{i,j}(u, \theta, k) &= \gamma_{i,j} + \alpha_{i,j}(k) + \log \text{SMPD}_i(u) \\
 &+ \beta_{i,j,1} \text{HighDev}_i(u) + \beta_{i,j,2} \text{Med.Dev}_i(u) \\
 &+ \beta_{i,j,3} \text{LowDev}_i(u) + \beta_{i,j,4} \text{OpenDev}_i(u) \\
 &+ \beta_{i,j,5} \text{LCH}_i(u) + \beta_{i,j,6} \text{HSI}(u) \\
 &+ \beta_{i,j,7} \left(\text{NDVI}_{i,j}(u) - \text{NDVI}_{\text{prev.sem}}(u) \right) + \beta_{i,j,8} \text{NDVI}_{i,j}(u) \\
 &+ \beta_{i,j,9} (\text{HighDev}_i(u))^2 + \beta_{i,j,10} (\text{Med.Dev}_i(u))^2 \\
 &+ \beta_{i,j,11} (\text{LowDev}_i(u))^2 + \beta_{i,j,12} (\text{OpenDev}_i(u))^2 \\
 &+ \beta_{i,j,13} (\text{LCH}_i(u))^2 + \beta_{i,j,14} (\text{HSI}(u))^2 \\
 &+ \beta_{i,j,15} \left(\text{NDVI}_{i,j}(u) - \text{NDVI}_{\text{prev.sem}}(u) \right)^2 + \beta_{i,j,16} \left(\text{NDVI}_{i,j}(u) \right)^2.
 \end{aligned} \tag{M3}$$

¹⁹ To accomplish the marginal nonparametric analysis for each covariate, we first create lower resolution ($2,500 \times 2,500$) versions of the covariates for computational ease and since our goal is exploratory analysis. Next, we use the `spatstat` function `rhohat.ppp` on the unmarked point pattern with the default estimation settings to obtain the nonparametric, smooth result. The plots of the results are in the supplementary material. It is important to note that our preliminary, marginal analysis is done for each covariate separately and does not account for other covariates included in the model; our goal was to inspect the basic relationship between the intensity of VF cases and the covariates individually.

Table 2

AIC scores for models M1, M2, and M3. Bold values indicate the model with minimum AIC in a given year and semester.

Year & Sem.	AIC: Offset Only (M1)	AIC: Log-linear (M2)	AIC: Log-quadratic (M3)
2013 1	1530.11	1532.34	1534.06
2013 2	1646.16	1644.10	1644.45
2014 1	2339.02	2331.76	2342.42
2014 2	1958.97	1949.45	1943.60
2015 1	2006.85	1995.51	2001.13
2015 2	4616.83	4580.28	4563.67
2016 1	4792.75	4720.43	4719.69
2016 2	4812.26	4755.41	4742.71
2017 1	4682.60	4628.47	4609.47
2017 2	4787.13	4742.22	4741.23
2018 1	5213.87	5152.18	5153.83
2018 2	4659.58	4586.92	4594.64
2019 1	4639.47	4560.10	4552.91
2019 2	5602.80	5525.78	5524.54
2020 1	5372.47	5311.74	5312.01
2020 2	6212.26	6119.80	6111.37
2021 1	6727.20	6660.68	6660.53
2021 2	6305.75	6246.62	6243.73
2022 1	5776.19	5691.47	5691.49
2022 2	5792.94	5742.79	5740.43
2023 1	5832.40	5792.28	5791.06
2023 2	6377.30	6325.62	6324.58

Note that SMPD estimates are used as an offset term to ensure that the dependence of VF cases on covariates is considered after accounting for the spatially varying at-risk population for VF.

The R package `spatstat` provides a robust toolkit for the analysis of spatial point patterns (Baddeley & Turner, 2005). Exploratory analysis, statistical inference on the covariates, and model validation were conducted mainly using tools from the `spatstat` package. We have used the `spatstat` function `ppm` to fit the models. We note that we have increased the resolution of the quadrature scheme used in model fitting from its default setting to a 250×250 grid²⁰ of dummy points. Furthermore, our model results involve randomization as we have used the Huang and Ogata model improvement procedure (HOI procedure; see (Huang & Ogata, 1999)) to obtain more accurate parameter estimates.²¹

2.4. Predicted and relative intensities

An advantage of using spatial point pattern techniques is that we obtain a prediction of the intensity of VF cases in all pixels from which we obtained quadrature points to fit the inhomogeneous Poisson process model. The predicted intensity of a point process model also provides the basis for spatial point pattern model validation (see (Baddeley et al., 2015), Chapter 11).

Relative intensity²² estimates based on predicted intensity are a simple way to show congruence between fitted spatial point process models and a smooth, nonparametric estimate of intensity of VF cases for a given semester. Let X_{ij} denote the unmarked spatial point process of VF cases (i.e. the superposition of all four types of VF cases, one for each age-sex category) in year i , semester j and let $\lambda_{ij}(u)$ denote the true intensity of X_{ij} at $u \in W$. Note that X_{ij} would be the unobserved process generating VF cases among the Medicaid population and the process for which we are interested in learning the intensity function. The relative intensity of X_{ij} and a fitted model for the intensity of X_{ij} is given by $r_{ij}(u) = \frac{\lambda_{ij}(u)}{\hat{\lambda}(u)}$ where $\hat{\lambda}_{ij}(u, \theta) := \sum_{k=1}^4 \hat{\lambda}_{ij}(u, \theta, k)$ denotes the estimated intensity function of X_{ij} . We estimate the relative intensity based on an observation of X_{ij} by

$$\hat{r}_{ij}(u) = \frac{1}{e(u)} \sum_{l=1}^n \frac{1}{\hat{\lambda}(x_l, \theta)} \kappa(u - x_l) \quad (1)$$

where x_l , $l = 1, \dots, n$ denotes the locations of n VF cases in an observation of X_{ij} , κ is a smoothing kernel, and $e(u)$ is an edge correction²³ of the estimate to deal with the bias arising from fact that we only observe the pattern inside the window W of

²⁰ While the entire grid is 250×250 dummy points, those points located where any covariate was labeled NA were removed from the quadrature scheme used to estimate the intensity over the window of study.

²¹ See documentation of the `ppm` function in `spatstat` for details.

²² See (Baddeley et al., 2015), chapter 11, section 11.2.1 for a description of this approach.

²³ We used the `spatstat` default adaptive bandwidth edge correction given by $e(u) = \int_W \kappa_h(u)(u - v) dv$ for $u \in W$. Here $h(u)$ denotes a spatially varying bandwidth; see section 2.3 in (Davies & Baddeley, 2018) for more details on edge correction factors in adaptive bandwidth kernel density estimation.

Table 5

Log-quadratic model (M3) results (part 1).

Year & Sem.	HI Dev. Land Cover	(HI Dev. Land Cover) ²	MI Dev. Land Cover	(MI Dev. Land Cover) ²
2013 1	−3.4824	11.3553	3.0049	−1.6123
2013 2	−5.4313	6.1955	5.5886	−4.7853
2014 1	−1.3192	−1.1825	−0.3991	1.2238
2014 2	0.0135	−7.9046	3.5433	−2.4806
2015 1	−3.1984	−4.7428	9.3583 (**)	−7.7489 (*)
2015 2	0.3220	−10.6806	7.5536 (***)	−7.6648 (***)
2016 1	−2.5286	−2.8460	2.8000	−1.8676
2016 2	1.1029	−13.0552	7.9381 (***)	−7.4005 (**)
2017 1	3.0055	−5.1836	2.8471	−0.6352
2017 2	−2.2079	3.4432	1.4116	0.4466
2018 1	−3.4040	0.1017	1.5197	−1.2436
2018 2	−0.6315	−15.2196 (*)	4.0837 (*)	−3.1023
2019 1	−2.5176	−6.3615	8.0048 (***)	−6.9803 (***)
2019 2	−1.1320	−9.8077	6.6170 (***)	−6.2169 (**)
2020 1	−1.3832	−11.1187	3.7983 (*)	−3.0377
2020 2	−0.7143	−7.3655	7.5295 (***)	−7.0084 (***)
2021 1	−1.4912	−6.5351	4.7413 (**)	−5.3110 (**)
2021 2	0.7418	−12.9373 (*)	2.5331	−3.1938
2022 1	0.4837	−7.1355	3.7277	−2.9860
2022 2	−1.5946	0.2336	4.5085 (*)	−3.0076
2023 1	−1.8311	−1.8202	6.4822 (***)	−5.8779 (**)
2023 2	−0.5544	−3.2980	4.7336 (**)	−3.5555
Year & Sem.	LI Dev. Land Cover	(LI Dev. Land Cover) ²	Open Dev. Land Cover	(Open Dev. Land Cover) ²
2013 1	−2.2963	3.8718	−1.2244	6.0560
2013 2	−5.8957	10.1984	−5.8532	21.9286 (*)
2014 1	7.5806	−11.8535	1.3906	5.7887
2014 2	−3.7812	9.3086	−5.0922	20.0917 (*)
2015 1	−6.5917	9.7044	−0.4611	13.5975
2015 2	−1.3079	2.3211	0.5390	4.3274
2016 1	6.6814 (*)	−10.0837 (*)	1.9083	−3.0759
2016 2	−3.2908	4.8748	0.4334	10.3790 (*)
2017 1	4.3294	−2.9084	2.7970	3.0712
2017 2	4.0877	−5.3628	2.4792	1.6494
2018 1	4.9871	−7.7329	0.5224	2.4936
2018 2	−1.8966	2.6986	1.2373	3.1046
2019 1	2.7984	−6.2454	1.6506	5.9389
2019 2	−1.6875	2.2532	0.9075	1.1761
2020 1	0.4808	−3.4750	0.0467	6.3098
2020 2	−3.4336	4.7989	2.5962	2.3152
2021 1	−1.4943	1.6201	1.8301	1.9898
2021 2	−0.3318	0.2337	0.4162	6.3458
2022 1	2.1512	−2.1166	1.5965	2.1320
2022 2	−1.5556	4.4215	1.5342	4.9747
2023 1	−1.0925	1.9155	0.7537	4.3415
2023 2	−1.0841	2.1273	2.1322	1.1552
Year & Sem.	Land Cover Change	(Land Cover Change) ²	NDVI	(NDVI) ²
2013 1	226.8914	−23514.4779	0.6008	−1.9025 (**)
2013 2	62.6800	−1536.4377	0.0397	−1.0720
2014 1	−28.9900	−191.7409	−0.4093	0.0344
2014 2	66.9085	−2229.7187	−0.2163	−0.8995
2015 1	36.7242	−1724.3147	−0.1126	−0.3097
2015 2	−24.9648	237.6961	0.0780	−0.3485
2016 1	−3.3203	70.3671	−0.2853	−0.1491
2016 2	43.1190	−1736.8326	0.1496	−0.6262 (*)
2017 1	70.3252 (*)	−2118.0934	−0.3356	−0.5593
2017 2	45.8926	−2321.0219	−0.1447	−0.4487
2018 1	19.1148	−93.1736	−0.1212	−0.3448
2018 2	21.3111	−282.7044	−0.5720 (***)	−0.1077
2019 1	8.9094	−92.2781	−0.4413	−0.0652
2019 2	8.1881	−20.2652	−0.4796 (**)	−0.1322
2020 1	11.6444	−251.6458	−0.5971 (**)	0.1478
2020 2	25.3726	−551.0562	−0.3260	−0.0065
2021 1	−12.3896	111.6324	−0.3435	−0.1932
2021 2	−20.7534	225.8370	−0.5385 (**)	−0.3074
2022 1	40.1725	−672.9225	0.0812	−0.3845
2022 2	2.0177	−245.6056	−0.1844	−0.9282 (**)
2023 1	19.8741	−581.2225	−0.2621	−0.2420
2023 2	−7.4730	88.8326	0.0484	−0.6841 (**)

Year & Sem.	NDVI – NDVI_lag	(NDVI – NDVI_lag) ²	HSI	(HSI) ²
2013 1	–0.4279	–2.7740	0.4251	0.1086
2013 2	4.1768	–14.7466 (*)	0.0842	–0.9306
2014 1	0.0811	–0.9882	0.3888 (*)	–0.3253
2014 2	2.3529	–16.0629	0.1588	–0.9128
2015 1	2.0511 (*)	–1.8780	0.1804	–0.1135
2015 2	–0.7594	–2.4949	0.3083 (**)	0.2672 (*)
2016 1	1.8493 (***)	–0.8607	0.4663 (***)	–0.0147
2016 2	–1.7628 (***)	–0.4581	0.3715 (**)	0.0267
2017 1	2.0747 (***)	0.1835	0.5954 (***)	0.2170
2017 2	–1.2603 (**)	0.9139	0.4684 (***)	0.1101
2018 1	1.6143 (**)	–1.3609	0.2960 (**)	0.1004
2018 2	–0.2450	–0.4421	0.4694 (***)	0.1021
2019 1	1.6823 (**)	1.3518	0.4481 (***)	0.1478
2019 2	–0.2541	0.8859	0.5254 (***)	–0.0213
2020 1	0.7734 (*)	0.8841 (*)	0.3854 (***)	0.1665
2020 2	–0.3992	0.6137	0.5456 (***)	–0.2078
2021 1	0.8382 (*)	0.6158	0.2975 (***)	0.0618
2021 2	–0.5155	–2.1473	0.2949 (**)	0.0267
2022 1	1.3215 (*)	–1.3828	0.4621 (***)	0.0233
2022 2	–2.1640 (***)	–1.0838	0.1318	0.0138
2023 1	1.0515 (*)	0.2893	0.3039 (**)	0.1068
2023 2	–0.7239	0.0062	0.2898 (***)	0.0662

finite area²⁴ (Baddeley & Turner, 2005). The relative intensity is simply a weighted estimate of the intensity of VF cases with the weight at each observed VF case equal to the inverse of the model-predicted intensity at that location (in our case we use predicted intensities of the parametric models M1, M2, and M3). We discuss predicted and relative intensity estimates based on our models in 3.4.

3. Model results

Table 2 displays Akaike Information Criterion (AIC) scores for models M1, M2, and M3 for all 22 semesters, and estimated coefficients for all three models appear in Tables 3–5 located in the Appendix (4). For visual inspection of the trend of the estimated coefficients for model M2 over time, we have also furnished Fig. 5 (since the model fits of M2 and M3 do not differ significantly based on AIC score in most semesters, we provide the plots only for M2 results). As expected, the log-quadratic (M3) outperforms the remaining models in more than half (14 of 22) of the semesters in terms of AIC, albeit by only a small margin (<3 in AIC score) relative to M2 in 8 of the 14 semesters where it outperformed the other two models. Notably, all but one semester (the first semester of 2013) show improved model fit compared to M1 when incorporating covariates in the model; the AIC differences in the first semester of 2013 are weak, with M1 and M2 differing by about 2.2 in AIC score. This may be due to that semester also having the smallest sample size of observed cases. AIC score differences between M1 and M2 are substantial (>7) for all but the first two semesters of the study period. Such results indicate that the addition of covariates measured at fine spatial resolutions are useful for modeling the intensity of VF cases for the Medicaid population.

Since we treat the residential address of VF cases as a marked point pattern, we can interpret the results of each model as providing four separate spatial point patterns based on age and sex category. By incorporating age-sex specific intercept terms, we are estimating an increase or decrease in the baseline intensity for that specific age-sex category with respect to the base category of males aged 18 – 44. Tables 3–5 demonstrate that, beyond the spatially varying estimate of Medicaid population, the intensity of VF cases differs significantly by age of the Medicaid enrollees. Those 45 years and older tend to have higher incidence of VF across both genders when compared to the intensity of Males 18 – 44 years of age. While we anticipated differences in VF case intensity for older Medicaid enrollees, we also expected to see a discrepancy between sexes which our results do not confirm. The time trend shown in the first two leftmost panels in the top row of Fig. 5 show a decreasing size in the effect of age on the intensity of VF cases among Medicaid enrollees. An identical pattern holds for the coefficients estimated for models M2 and M1. These downward trends may be related to the similar downward trend in the percentages of total cases made up of Medicaid enrollees of age 45 years or older that are readily apparent in Fig. 4. Changes in Medicaid enrollment patterns and/or increases in the common sources of VF risk among older and younger enrollees are also possible explanations for the trend.

3.1. HSI results

Since model M3 only substantially outperforms model M2 (AIC score >7) in about five semesters and since, for most estimates in model M3, the quadratic terms are statistically insignificant, we focus on interpreting the results of model M2.

²⁴ see chapter 6, section 6.5.1 and chapter 11, section 11.2.1 in (Baddeley & Turner, 2005) for more details on edge correction factors and relative intensity estimates for model validation, respectively.



Fig. 5. Time series of coefficient estimates from model M2 for all semesters (red lines). The secondary vertical axis pertains to the average NDVI (green curve) over the study window. Dark red points indicate semesters where the coefficient was significant at the 95% level or greater (see Table 4). Since estimates for land cover change were large in magnitude, changed dramatically between semesters and were statistically insignificant in most semesters, we do not include them in the plot.

Moreover, for HSI, the linear terms in the log-quadratic model, M3 remain statistically significant at the 95% level or greater and the magnitude of the coefficients themselves are similar to those of the log-linear model, M2 in all but the first semester of 2013 (see Tables 5 and 4).

Recall that we have standardized the HSI and NDVI covariates and then smoothed the standardized result. Fig. 5 reveals a relatively stable pattern of coefficient estimates over time. The largest observed effect for HSI occurs in the first semester of 2017 where a one-standard deviation²⁵ increase in HSI in the (roughly 1-kilometer) neighborhood of a point is associated with an increase in the intensity of VF cases by a factor of $e^{\hat{\beta}_{2013,1.6}} = e^{0.586} \approx 1.797$, i.e. the intensity is expected to increase by about 80% when holding all other covariates constant. In the second semester of 2022, this increase shrinks to only 17%. Averaging the estimated effects for all 22 semesters reveals that, on average, a one-standard deviation increase in HSI in the neighborhood of a point is associated with an increase in the intensity of VF cases by 50% when holding other covariates constant. We find statistically significant evidence that HSI has a positive effect on the intensity of VF cases among Medicaid patients in 18 of the 22 semesters, revealing a consistent pattern in the utility of soil habitat suitability as a measure of VF risk.

3.2. NDVI results

Current semester NDVI has a consistently negative effect on the intensity of VF cases. The largest effect in magnitude for NDVI is observed in the second semester 2018, where an increase by a single standard deviation (computed from all of the pixels of the study window) in the neighborhood of a point, is associated with a decrease in the intensity of VF cases by a factor of $e^{\hat{\beta}_{2018,2.8}} = e^{-0.604} \approx 0.547$, or a decrease by about 45%.

The inverse relationship between the intensity of VF cases and NDVI may be due to higher soil moisture associated with larger NDVI values. A similar inverse relationship between the intensity of VF cases (incidence) and higher concurrent NDVI is seen in Fig. 3 of (Stacy et al., 2012), where the authors note that a potential reason for the relationship is that dust, which can carry arthroconidia, is inhibited by moist soil. Since our case and NDVI data are aggregated over the full semester, we cannot directly compare our results with those of (Stacy et al., 2012). However, the changing signs for the estimated coefficients for NDVI difference is potentially related to the dynamic effect of NDVI on the intensity of VF cases as discussed in (Stacy et al., 2012); the authors note that higher NDVI measured up to (even slightly over a year) prior can have a positive effect on the intensity of VF cases (see the middle panel in the second row of Fig. 5 where, for reference, we also plot the global, spatial average NDVI by semester).

3.3. Land cover and land cover change results

For the exposure to land cover type and land cover change, we note that it is helpful to consider the interpretation of the estimated effects by considering an example (since we do not standardize these covariates). For example, a given residence highly exposed to open land development in the first semester of 2023 might have an associated pixel value of 0.25 (rather than 1 due to the smoothing). In this case, the estimated intensity of VF cases at a residence where the land cover change covariate is 0.25 is increased by a factor of approximately $e^{\hat{\beta}_{2023,1.4} * 0.25} = e^{2.615 * 0.25} = 1.923$ or by 92% compared to places where there is relatively little nearby open land development. Moreover, this effect decays rapidly over space.

Interestingly, the estimated effects of exposure to high-intensity land development are consistently negative and, especially in the second half of the study period, statistically significant. While high-intensity development includes residential real estate where population density is high, it also includes commercial/industrial properties which are not likely to be locations where VF cases are observed. Importantly, places with high-intensity development have mostly impervious surfaces (see (Multi-Resolution Land Characteristics (MRLC) Consortium n.d.)) where uncovered soil is less likely to be observed. At the same time, we observe strong evidence of positive effects for exposure to medium and low-intensity development in most semesters and relatively larger positive effects for exposure open development. Furthermore, the estimated effects for each of these covariates are statistically significant at the 95% level or greater in at least 14 of the 22 semesters. Taken together, these results suggest that both the soil habitability (HSI) and the degree of uncovered soil exposure are useful for understanding VF risk among the Medicaid population.

Treating land cover change as a continuous covariate allows us to estimate the diffusive effect of exposure to arthroconidia arising from soil disruption due to new development. Our approach captures the effect of distance to land cover change as well as the quantity of new development in the vicinity of VF cases. The relatively large coefficients reported in Tables 3–5 are due to the fact that the land cover change variable can theoretically take values from 0 (no new development effects) to 1 (new development effects recorded in a 30-meter pixel). However, actual pixel values after applying Gaussian smoothing to the binary rasters are much smaller. Notably, the estimated effects of land cover change are not statistically significant at the 95% level for all but a single semester. The lack of a consistent, positive effect as we expected is possibly due to the interaction between land cover change and the land cover data that is also included as smoothed exposure variables. Our results match those of (Porter et al., 2024), who also found no significant effect of land cover change on VF incidence.

²⁵ The standard deviation of the entire HSI raster used for analysis is 0.191 over the study window, while the mean HSI is about 0.401.

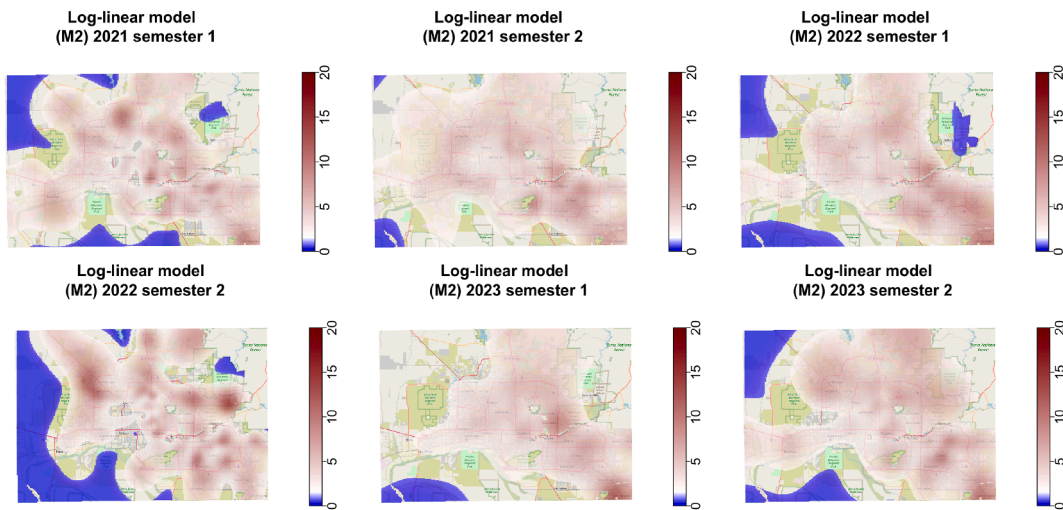


Fig. 6. Estimated relative intensities for M2 in the last six semesters of the study period. Red indicates areas where the model underestimates the intensity of VF cases with darker red showing more severe underestimation, while blue indicates areas where the intensity is overestimated. Relative intensities that were NA, and where the model roughly conformed to the empirical intensity (i.e. where the relative intensity was below 2 and above 0.5) are not shown in color. To highlight general trends in model performance, pixels with predicted intensity values (which are used to calculate the weights in (1)) less than 0.001 are replaced with the minimum predicted intensity greater than or equal to 0.001. **Source:** AHCCCS, 2023. CHiR created patient record files from Arizona Medicaid (AHCCCS) data that were used for this analysis. The basemap layer was created using the R package basemaps (Schwalb-Willmann, 2024).

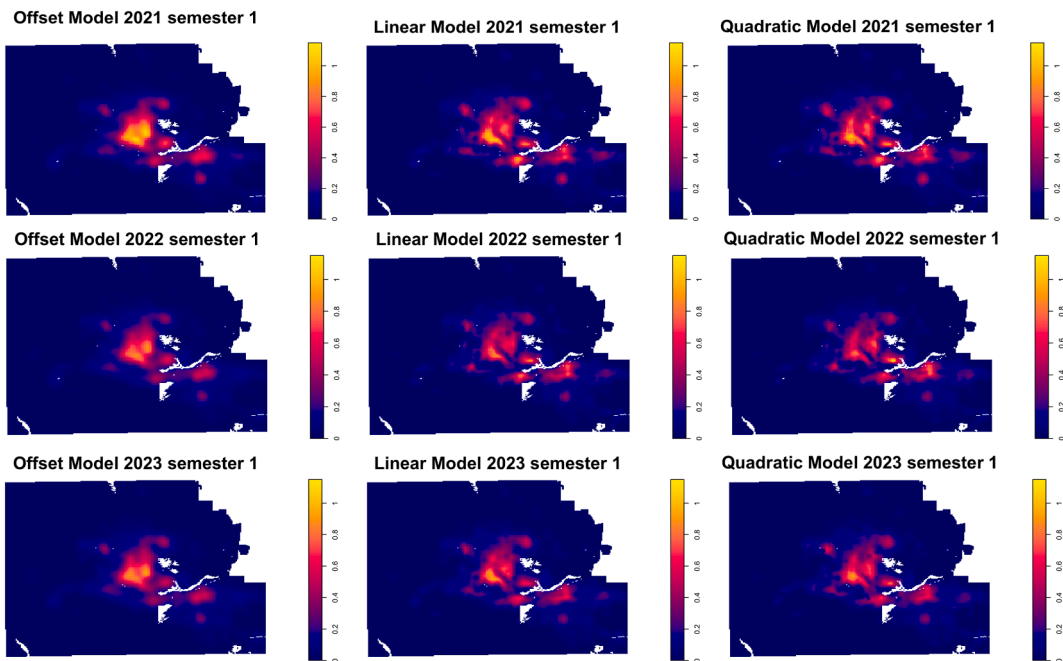


Fig. 7. Predicted intensities for the first semester of the last three years in the study period. **Source:** AHCCCS, 2023. CHiR created patient record files from Arizona Medicaid (AHCCCS) data that were used for this analysis.

3.4. Predicted and relative intensity results

We sum up the intensity patterns for all four age and sex categories to obtain a full predicted intensity surface to estimate the intensity of VF cases in areas where there are no observed cases for a given semester based on the covariates of interest.

In Fig. 7 we display the predicted intensities for models M1, M2, and M3 for the first semester of 2021, 2022, and 2023 to give insight into the temporal evolution of the predicted intensities²⁹. While the predicted intensities for models M2 and M3 do not appear differ substantially, the difference between these two and the more smoothly varying predicted intensity of model M1 is visually apparent. This, along with the better fit of models M2 and M3 in terms of AIC suggest that the covariates provide greater insight into the intensity of VF cases for Medicaid patients beyond the general trend of cases that follows the Medicaid population density captured by the offset term. The sharper peaks and valleys reveal a more spatially refined predicted intensity for the models with covariates included. The predicted intensity of a point process model also provide the basis for spatial point pattern model validation (see (Baddeley et al., 2015), Chapter 11).

All models produced relatively small predicted intensities at various observed VF case locations, creating large outliers in the weights used for computing (1). We limit the visual effect of outliers in the relative intensity plots shown in Fig. 6 to emphasize the general trend in model performance over the study window. To achieve this, when computing the weights in (1), we replaced predicted intensities less than 0.001, with the smallest predicted intensity greater than or equal to 0.001. Using an adaptive-bandwidth kernel²⁶ allows us to tailor our relative intensity estimate to protect privacy while highlighting regions within the study window where the models fail to explain the observed intensity of VF cases.²⁷

Fig. 6 shows the estimated relative intensity for the last six semesters of the study period. The evolving pattern of underestimation highlights that while the covariates improve model fit in terms of AIC, there are clear points where the model fit breaks down and fails to capture certain hotspots in the intensity of VF cases. An important observation is that many of the more pronounced hotspots of underestimated intensity appear as one moves away from the center of the study window where model M2 predicts fewer cases (see the top two panels of the middle column of Fig. 7). Relative intensity plots for all 22 semesters and all three models are available in the supplementary material.

4. Discussion and conclusion

Our results suggest that for Medicaid enrollees in Arizona, local environmental factors around one's residence may influence risk of Valley fever. Indeed, we have shown that such factors that are known to influence the intensity of VF cases based on spatially aggregated data can inform estimates of VF risk at smaller scales. Our analysis reveals an inverse relationship between NDVI and observed VF cases at a local spatial scale and suggests that lagged NDVI effects may be useful for understanding VF risk among Medicaid patients. We have also shown that both soil habitat suitability of the *Coccidioides* fungus and exposure to different land cover types can influence VF risk at the local spatial scale. Lastly, our results are consistent with other studies that find land cover change does not have an effect on the intensity of VF cases (see e.g. (Porter et al., 2024) for an example where similar land cover change data are used and (Blair et al., 2014) for a specific example regarding employees near a construction/new development site). However, the semester structure of our data likely limits our ability to detect the effects from changes in covariates.

While our approach provides evidence at the spatial point level for environmental covariate effects on the intensity of VF cases, even our best model substantially under/overestimates VF case intensity (Fig. 6), especially in areas at the boundary of the more densely populated region of the study window. There are multiple possible reasons for this, among which the most likely is that we are using only estimates of the spatially varying Medicaid population. Indeed, some of the census block groups used for the estimates have relatively large margins of error.

Our results demonstrate that geocoded residential addresses of VF patients can be useful for estimating VF risk at the residential level. However, more sophisticated spatial point process methods could help identify localized outbreaks. For example, a model that incorporates unobserved spatial 'random effects' could formally identify the regions in the study window where the covariates fail to effectively capture the spatial dynamics of the intensity of VF cases.²⁸

Our exploratory analysis in this paper has highlighted the utility of spatial point pattern analysis techniques in understanding the intensity of VF cases; however, since our data is temporally aggregated to semesters, this poses similar challenges to the spatial aggregation of VF data. If the VF cases were aggregated at, say, monthly intervals, we would observe different spatial point patterns which may reveal effect sizes that differ from those estimated at the semester-based temporal scale. Temporal aggregation of covariate data also plays a critical role in analyzing VF case intensity. It has been shown, for example, that one-month lagged precipitation is useful for predicting the intensity of VF cases at the county level (Kollath et al., 2022). Indeed, covariates that may be used to identify short-term spatially localized effects that can influence the intensity of VF cases (e.g. weather events) are not included here due to the aggregation of the case data. To address this, higher frequency environmental covariate data could be exploited in a spatiotemporal point pattern model for VF cases identified by location as well as date of diagnosis or first symptoms.

²⁹ Predicted intensity plots from all three models and for all semesters is available in the supplementary material.

²⁶ See (Baddeley & Turner, 2005) and (Baddeley et al., 2015), especially chapter 6, for details on nonparametric intensity estimation.

²⁷ A global bandwidth of 15 km was chosen for use as a pilot bandwidth which is passed to the `spatstat` function `bw.abram.ppp`. See also (Abramson, 1982) and (Davies & Baddeley, 2018) for details on how this method works.

²⁸ See (Baddeley et al., 2015), chapter 12 for a discussion of relevant methods (e.g. Cox processes).

CRediT authorship contribution statement

John Ginos: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Beckett Sterner:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Petar Jevtić:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used OpenAI's ChatGPT (GPT-4o and GPT 5.2) in order to prepare and format the bibliography as well as certain presented figures and tables in LaTeX. The authors also used ChatGPT (GPT-4o) to generate ideas for the visualization of NDVI data (e.g. using a normalized difference between two NDVI rasters) and (mostly refine) small portions of the R code used for the data analysis medium. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Conflict of interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.idm.2026.03.009>.

Appendix

In all of the tables below, we use***, and ** to denote the 95%, 99%, and 99.9% significance level for the Wald test for model coefficients (see (Baddeley et al., 2015) page 374).

Table 3
Offset only model (M1) results

Year & Sem.	Intercept	Females 45+	Males 45+	Female 18–44
2013 1	2.7743 (***)	1.2640 (***)	1.0774 (***)	0.0531
2013 2	2.9718 (***)	0.9933 (***)	1.0733 (***)	−0.0107
2014 1	3.3425 (***)	1.1029 (***)	1.1791 (***)	−0.0561
2014 2	3.1986 (***)	0.7800 (**)	1.2443 (***)	0.0821
2015 1	3.2836 (***)	0.8488 (***)	1.0221 (***)	−0.0563
2015 2	4.1586 (***)	1.1980 (***)	1.2599 (***)	−0.1794
2016 1	4.4357 (***)	0.9868 (***)	0.7981 (***)	−0.1817
2016 2	4.2812 (***)	0.9735 (***)	1.0276 (***)	0.1031
2017 1	4.2920 (***)	1.0379 (***)	1.1040 (***)	−0.3195
2017 2	4.4856 (***)	0.8059 (***)	0.8575 (***)	−0.3990 (*)
2018 1	4.4325 (***)	0.9962 (***)	0.8786 (***)	0.0831
2018 2	4.4665 (***)	0.7189 (***)	0.8042 (***)	−0.1810
2019 1	4.3959 (***)	0.8201 (***)	0.8208 (***)	0.0135
2019 2	4.7617 (***)	0.7497 (***)	0.5992 (***)	−0.0640
2020 1	4.6691 (***)	0.6127 (***)	0.6711 (***)	0.1294
2020 2	5.0389 (***)	0.4441 (***)	0.4567 (***)	−0.2105
2021 1	5.0016 (***)	0.6211 (***)	0.5869 (***)	−0.1367
2021 2	4.8239 (***)	0.7228 (***)	0.7297 (***)	0.0293
2022 1	4.8651 (***)	0.6302 (***)	0.4112 (***)	−0.1628
2022 2	4.8201 (***)	0.6441 (***)	0.5729 (***)	−0.0276
2023 1	4.8284 (***)	0.6522 (***)	0.4992 (***)	0.0722
2023 2	5.1164 (***)	0.2772 (**)	0.5164 (***)	−0.1888

Table 4
Log-linear model (M2) results

Year & Sem.	Intercept	Females 45+	Males 45+	Female 18–44
2013 1	2.0229 (***)	1.3269 (***)	1.1650 (***)	0.1452
2013 2	1.9076 (**)	0.9800 (**)	1.0797 (***)	0.0040
2014 1	2.6615 (***)	1.0800 (***)	1.1399 (***)	−0.0448
2014 2	1.9603 (***)	0.7594 (**)	1.2232 (***)	0.0279
2015 1	2.2596 (***)	0.8823 (***)	1.0802 (***)	0.0510
2015 2	3.2487 (***)	1.2160 (***)	1.2527 (***)	−0.1497
2016 1	3.6708 (***)	0.9923 (***)	0.7874 (***)	−0.1916
2016 2	3.3244 (***)	0.9679 (***)	1.0234 (***)	0.0754
2017 1	3.6070 (***)	1.0070 (***)	1.0780 (***)	−0.3665 (*)
2017 2	4.0649 (***)	0.8027 (***)	0.8173 (***)	−0.3827 (*)
2018 1	3.7853 (***)	1.0395 (***)	0.9205 (***)	0.1238
2018 2	3.7261 (***)	0.7874 (***)	0.8305 (***)	−0.0938
2019 1	3.3696 (***)	0.8316 (***)	0.8427 (***)	0.0144
2019 2	3.9815 (***)	0.7057 (***)	0.5742 (***)	−0.0628
2020 1	4.3160 (***)	0.6135 (***)	0.6404 (***)	0.1256
2020 2	4.0528 (***)	0.4267 (***)	0.4523 (***)	−0.2222
2021 1	4.6156 (***)	0.6145 (***)	0.5560 (***)	−0.1760
2021 2	4.4658 (***)	0.6852 (***)	0.7263 (***)	0.0116
2022 1	3.6008 (***)	0.5898 (***)	0.3826 (***)	−0.1691
2022 2	3.5341 (***)	0.6279 (***)	0.5606 (***)	−0.0553
2023 1	4.0754 (***)	0.6676 (***)	0.5194 (***)	0.0561
2023 2	4.1453 (***)	0.3067 (***)	0.5433 (***)	−0.1951

Year & Sem.	HI Dev. Land Cover	MI Dev. Land Cover	LI Dev. Land Cover	Opne Dev. Land Cover
2013 1	−2.0914	1.6984 (*)	0.1682	2.3698
2013 2	−2.8474 (*)	1.9733 (**)	1.1365	1.3289
2014 1	−1.7291	1.0575 (*)	1.0443	3.6743 (*)
2014 2	−2.5258 (*)	1.9655 (**)	2.2315 (*)	1.5126
2015 1	−3.1121 (*)	2.2037 (***)	0.3764	4.2759 (*)
2015 2	−1.5400	1.4333 (***)	2.0264 (***)	2.6273 (*)
2016 1	−3.4600 (***)	1.5651 (***)	1.8984 (***)	1.7968
2016 2	−1.6075	1.9035 (***)	1.2496 (*)	4.3399 (***)
2017 1	−1.4403	1.8141 (***)	−0.2135 (*)	4.3517 (***)
2017 2	−2.8228 (***)	1.6258 (***)	−0.3787 (**)	3.7185 (**)
2018 1	−3.2945 (***)	1.0311 (**)	1.2328 (**)	2.2627 (*)
2018 2	−4.3379 (***)	1.6237 (***)	0.5249	1.9747
2019 1	−3.3891 (***)	1.9204 (***)	1.1077	3.7321 (***)
2019 2	−2.9997 (***)	1.1712 (***)	1.1721 (**)	1.1956
2020 1	−3.5819 (***)	1.2787 (***)	−0.4945	2.1614 (*)
2020 2	−1.7816 (*)	1.3472 (***)	0.9070	3.5990 (***)
2021 1	−2.4279 (***)	0.5347 (*)	0.9903 (**)	2.6793 (***)
2021 2	−2.5117 (***)	0.4902	0.8317 (*)	2.7521 (***)
2022 1	−1.7351 (*)	1.7643 (***)	1.9646 (***)	3.1306 (***)
2022 2	−1.9951 (**)	2.0500 (***)	1.6341 (**)	3.2596 (***)
2023 1	−1.7833 (**)	1.2459 (**)	1.1244 (*)	2.6151 (***)
2023 2	−1.7771 (*)	1.8899 (***)	1.0569 (*)	2.6386 (**)

Year & Sem.	Land Cover Change	NDVI	NDVI–NDVI_lag	HSI
2013 1	−15.8411	−0.0483	0.1215	0.5119 (**)
2013 2	25.4965	−0.5157	0.3267	0.2216
2014 1	−18.7431	−0.4642 (*)	0.4351	0.4320 (**)
2014 2	9.3020	−0.5531 (*)	−0.6860	0.2995
2015 1	−2.9469	−0.3282	1.4534 (*)	0.1861
2015 2	−16.1428	−0.2324	−0.8838 (*)	0.3550 (**)
2016 1	−1.5790	−0.4456 (**)	1.6557 (***)	0.4796 (***)
2016 2	−2.6558	−0.2682 (*)	−1.4165 (***)	0.4303 (***)
2017 1	1.5107	−0.3481 (*)	0.7292 (*)	0.5861 (***)
2017 2	−11.4447	−0.3119 (*)	−0.5063	0.4945 (***)
2018 1	15.9109	−0.3717 (**)	1.1201 (**)	0.3022 (**)
2018 2	12.2335	−0.6042 (***)	−0.1393	0.4897 (***)
2019 1	1.9665	−0.5594 (**)	1.0097 (*)	0.5128 (***)
2019 2	6.2952	−0.5406 (**)	0.4918	0.5514 (***)
2020 1	3.1309	−0.5941 (***)	0.3856	0.4436 (***)
2020 2	9.3371	−0.4229 (**)	0.2182	0.5283 (***)
2021 1	−3.2843	−0.3927 (**)	0.7140	0.3215 (***)
2021 2	−12.2832	−0.5889 (***)	0.0802	0.3455 (***)

Table 4 (continued)

Year & Sem.	Land Cover Change	NDVI	NDVI–NDVI_lag	HSI
2022 1	24.9804 (*)	−0.0883	1.2620 (*)	0.4966 (***)
2022 2	−7.9355	−0.3499 (*)	−1.7181 (***)	0.1580
2023 1	−4.2403	−0.2537	0.5941	0.3491 (***)
2023 2	−4.4910	−0.0867	−0.5151	0.3268 (***)

Year & Sem.	Intercept	Females 45+	Males 45+	Female 18–44
2013 1	2.5736 (***)	1.2677 (***)	1.0956 (**)	0.1334
2013 2	2.9725 (***)	1.0075 (**)	1.1004 (***)	0.0687
2014 1	2.4532 (***)	1.0485 (***)	1.0886 (***)	−0.0827
2014 2	3.0887 (***)	0.7731 (**)	1.2103 (***)	0.0075
2015 1	2.3863 (**)	0.8983 (***)	1.0537 (***)	−0.0306
2015 2	3.1489 (***)	1.2243 (***)	1.2923 (***)	−0.1864
2016 1	3.1753 (***)	1.0146 (***)	0.7944 (***)	−0.1969
2016 2	3.3311 (***)	0.9898 (***)	1.0535 (***)	0.1258
2017 1	2.6140 (***)	1.0459 (***)	1.1124 (***)	−0.3257
2017 2	3.3790 (***)	0.8279 (***)	0.8828 (***)	−0.3839 (*)
2018 1	3.6758 (***)	1.0076 (***)	0.8749 (***)	0.0732
2018 2	3.6742 (***)	0.7558 (***)	0.8377 (***)	−0.1320
2019 1	2.4482 (***)	0.8641 (***)	0.8408 (***)	0.0207
2019 2	3.6423 (***)	0.7288 (***)	0.6123 (***)	−0.0495
2020 1	3.8225 (***)	0.6341 (***)	0.6704 (***)	0.1504
2020 2	3.8601 (***)	0.4229 (***)	0.4319 (***)	−0.2542 (*)
2021 1	4.5735 (***)	0.5831 (***)	0.5544 (***)	−0.1793
2021 2	4.6556 (***)	0.6788 (***)	0.6977 (***)	0.0010
2022 1	3.6288 (***)	0.6048 (***)	0.3934 (***)	−0.1674
2022 2	3.8284 (***)	0.6409 (***)	0.5655 (***)	−0.0555
2023 1	3.7352 (***)	0.6910 (***)	0.5447 (***)	0.1000
2023 2	4.2103 (***)	0.3036 (**)	0.5381 (***)	−0.2051

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